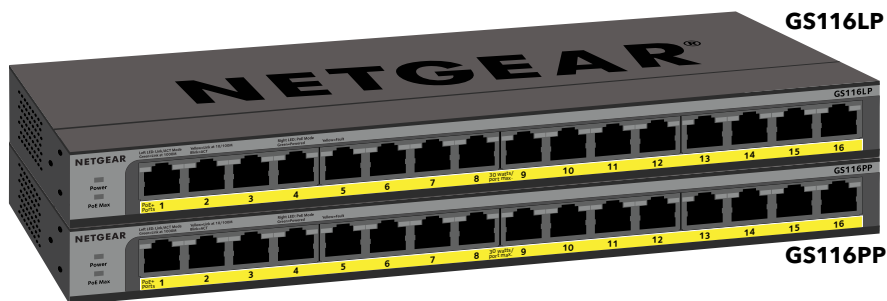


Installation Guide

16-Port Gigabit Ethernet PoE+ Unmanaged Switch (76W), Model GS116LP

16-Port Gigabit Ethernet PoE+ Unmanaged Switch (183W), Model GS116PP



Package contents

- Switch
- Power adapter
- Power cord (varies by region)
- Wall installation kit
- Rubber feet
- Rack-mount kit
- Mounting ties (for power adapter)
- Installation guide

1. Register the switch

Registration is required for warranty activation and support. For more information, visit netgear.com/about/warranty.

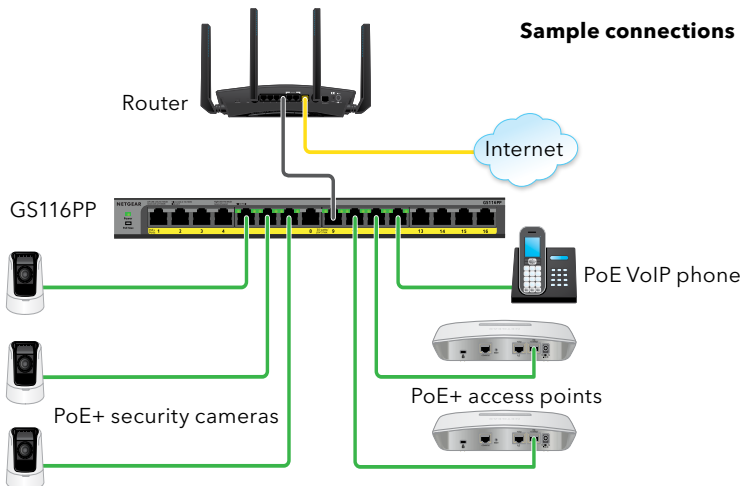
1. From a computer or mobile device that is connected to the Internet, visit my.netgear.com.
2. Log in to your NETGEAR account.

NOTE: If you don't have a free NETGEAR account, you can create one. The Your Registered Products page displays.

3. Click the **REGISTER NEW PRODUCT** button.
4. In the **SERIAL NUMBER** field, type the serial number of your switch. The serial number is 13 digits long. It is printed on the switch label.
5. From the **PURCHASE DATE** menus, select the date that you purchased the switch.
6. Click the **REGISTER** button.

Your switch is registered to your NETGEAR account. A confirmation email is sent to your NETGEAR account email address.

2. Connect the switch



1. Connect network devices to the ports on the switch.
2. Connect an RJ-45 port on the switch to a network.

NOTE: In a small office or home office network, connect the switch to the LAN port of a router that, in turn, is connected to a modem.

3. Power on the switch.

This switch is designed for indoor use only. If you want to connect it to a device located outdoors, the outdoor device must be properly grounded and surge protected, and you must install an Ethernet surge protector inline between the switch and the outdoor device. Failure to do so can damage the switch.

 **WARNING:** Before connecting this switch to outdoor cables or devices, see <https://kb.netgear.com/000057103> for safety and warranty information.

3. Check the LEDs

When you connect the power cord to the switch and plug it into an electrical outlet, the LEDs indicate the status:

LED	Description
Power	 Solid green: The switch is powered on and operating normally.  Off: Power is not supplied to the switch.
PoE Max (The status of the switch's PoE budget.)	 Off: Sufficient (more than 7W of) PoE power is available.  Solid yellow: Less than 7W of PoE power is available.  Blinking yellow: At least once during the previous two minutes, less than 7W of PoE power was available.
Left port LED	 Solid green: 1000 Mbps link on this port.  Blinking green: 1000 Mbps activity on this port.  Solid yellow: 100 Mbps or 10 Mbps link on this port.  Blinking yellow: 100 Mbps or 10 Mbps activity on this port.  Off: No link is detected on this port.
Right Port LED	 Solid Green: The port is delivering PoE power.  Off: The port is not delivering PoE power.  Solid yellow: A PoE fault occurred.

PoE considerations

The switch can supply up to 30W PoE+ (IEEE 802.3at) to each port:

- **Model GS116LP:** Supports a maximum PoE power budget of total 76W with its default 90W power adapter across all active PoE+ ports.
- **Model GS116PP:** Supports a maximum PoE power budget of total 183W with its default 200W power adapter across all active PoE+ ports.

The switch prioritizes the PoE+ power that it supplies in ascending port order (from port 1 to port 16). If the aggregate power requirements of all attached power devices (PD) exceed the power budget of the switch, the PD on the highest-numbered port is disabled to make sure that the PDs that are connected to the higher-priority, lower-numbered ports are supported first.

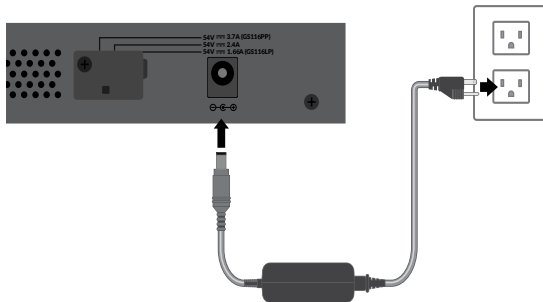
The following table shows the standard power ranges, calculated with the maximum cable length of 328 feet (100 meters). If a device receives insufficient PoE power from the switch, consider using a shorter cable.

Device Class	Compatible PoE Standard	Class Description	Maximum Power Supplied by the Switch	Power Delivered to the Device
0	PoE and PoE+	Default power (full)	15.4W	0.44W-12.95W
1	PoE and PoE+	Very low power	4.0W	0.44W-3.84W
2	PoE and PoE+	Low power	7.0W	3.84W-6.49W
3	PoE and PoE+	Mid power	15.4W	6.49W-12.95W
4	PoE+	High power	30.0W	12.95W-25.5W

Change the switch's flexible PoE budget

As an option, you can move the slider on the back of the switch to increase or decrease the PoE budget. You can increase the PoE budget if you purchase a higher wattage power adapter. You also can move the slider to a PoE budget that is lower than the power adapter wattage. This reduces your switch's power consumption. However, we recommend using the slider setting that matches your power adapter.

1. Power down the switch and disconnect the power cord.
2. Use a screwdriver to remove the metal plate that covers the slider.
3. Move the slider to the setting that corresponds to the wattage of the new power adapter.
4. Replace the metal plate.
5. Connect the power cord and power on the switch.



PoE troubleshooting

Here are some tips for correcting PoE problems that might occur:

- If the PoE Max LED is yellow, disconnect one or more PoE devices to prevent PoE oversubscription. Start by disconnecting the device from the highest-numbered port.
- For each powered device (PD) that is connected to the switch, the associated right port LED on the switch lights solid green. If the right port LED lights yellow, a PoE fault occurred because of one of the conditions listed in the following table.

PoE Fault Condition	Possible Solution
A PoE-related short circuit occurred on the port.	
The PoE power demand of the PD exceeded the maximum level that the switch permits. The maximum level is 15.4W for a PoE connection or 30W for a PoE+ connection	The problem is most likely with the attached PD. Check the condition of the PD, or restart the PD by disconnecting and reconnecting the PD.
The PoE current on the port exceeded the classification limit of the PD.	
The PoE voltage of the port is outside the range that the switch permits	Power cycle the switch to see if the condition resolves itself.

Mount the switch on a wall

We recommend that you use the wall-mount screws that are included. The screws are 0.25 in. (6.5 mm) in diameter, 0.86 in. (22 mm) in length.

1. Locate the two mounting holes on the bottom panel of the switch.
2. Mark and drill two mounting holes in the wall where you want to mount the switch.
The two mounting holes must be 4.27 in. (108.5 mm) apart, center-to-center.
3. Insert the supplied anchors into the wall and tighten the screws with a No. 2 Phillips screwdriver.
Leave about 0.15 in. (4 mm) of each screw protruding from the wall so that you can insert the screws into the holes on the bottom panel.

Install the Switch in a Rack

To install the switch in a rack, you need the rack-mount brackets and screws that are supplied with the switch.

1. Attach the supplied mounting brackets to the side of the switch.
Insert the screws provided in the product package through each bracket and into the bracket mounting holes in the switch.

2. Tighten the screws with a No. 2 Phillips screwdriver to secure each bracket.
3. Align the mounting holes in the brackets with the holes in the rack, and insert two pan-head screws with nylon washers through each bracket and into the rack.
4. Tighten the screws with a No. 2 Phillips screwdriver to secure mounting brackets to the rack.

Technical specifications

Specification	Description
Network interfaces	16 Gigabit Ethernet RJ-45 PoE+ ports that support 1 Gbps, 100 Mbps, and 10 Mbps
Network cable	Category 5e (Cat 5e) or higher rated Ethernet cable
Maximum PoE budget	
Model GS116LP	90W default adapter: 76W
Model GS116PP	200W default adapter: 183W
Optional power adapter	130W optional adapter: 115W
DC input power	
Model GS116LP	Input 54V, 1.66A (90W default adapter)
Model GS116PP	Input 54V, 3.7A (200W default adapter)
Optional power adapter	Input 54V, 2.4A (130W optional adapter)
Dimensions (W x D x H)	11.3 x 4.0 x 1.0 in. (286 x 102 x 27 mm)
Weight	1.94 lb (0.88 kg)
Operating temperature	32-104°F (0-40°C)
Operating humidity	10%-90% relative humidity, noncondensing
Compliance	FCC class A, CB, CE class A, VCCI class A, RCM class A, KC, BSMI, EAC

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(If this product is sold in Canada, you can access this document in Canadian French at <https://www.netgear.com/support/download/>.)

For regulatory compliance information including the EU Declaration of Conformity, visit <https://www.netgear.com/about/regulatory/>.

See the regulatory compliance document before connecting the power supply.

For NETGEAR's Privacy Policy, visit <https://www.netgear.com/about/privacy-policy>.

By using this device, you are agreeing to NETGEAR's Terms and Conditions at <https://www.netgear.com/about/terms-and-conditions>. If you do not agree, return the device to your place of purchase within your return period.

Do not use this device outdoors. The PoE source is intended for intra building connection only.

Applicable to 6 GHz devices only: Only use the device indoors. The operation of 6 GHz devices is prohibited on oil platforms, cars, trains, boats, and aircraft, except that operation of this device is permitted in large aircraft while flying above 10,000 feet. Operation of transmitters in the 5.925-7.125 GHz band is prohibited for control of or communications with unmanned aircraft systems.

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